

Solar Energy Conversion to Chemicals by Bacteria

In chemical production, there are different production mechanisms that are currently used to effectively create chemicals. There are systems that take advantage of the molecular engines bacteria possess and with manipulation, they are a small factory to produce a product of interest. Atmospheric carbon dioxide, CO₂, are at dangerously high levels and this detriment is only perpetuated by dependence on fossil fuel usage that continues to produce this harmful gas. There are a plethora of options available as alternatives to indirectly decrease our dependency on this source of energy such as renewable sources, such as wind, solar, and hydropower. Though development of these alternate resources provide alternatives, there is still enough of a dependency on CO₂ emitting that continues to lead to record high levels of emission. This leads to the purpose of this paper, to introduce bacteria to directly take CO₂ out of the air and convert, via fixation, the gas into a usable product. This would be vital as a direct solution to the high levels of CO₂ production where the purpose is to, on a large scale, recycle atmospheric CO₂ and reduce the negative effects of CO₂ pollution that include ocean acidification. Looking at bacterial manipulation of CO₂, bacterial metabolism often possess the ability to use energy from the sun to decrease costs of CO₂ from thousands of dollars' worth of reaction dependent upon catalysts. In recent advancements, manipulation of molecular CO₂ fixation had led to one of the most efficient and cost effective method of reducing atmospheric CO₂.

The question that you are asking as a reader may be, "But why haven't we used this method for CO₂ utilization sooner?!" As simple of an idea as this may sound, there are a few limitations in solar energy conversion. Of course, we have solar technology, such as solar panels, that are able to extract quite a bit of energy from solar light but they fail to actually store the energy effectively. Much like the solar panel, there is a lot of product that is being produced from solar energy, where the panels produce enough energy for a house, in our case a bacterium can produce chemicals such CH₃COOH, acetic acid but the issue was the utilization of the chemical by the organism. Previous mechanisms involve a photosynthetic organism, such as *Clostridium acetobutylicum*, that produces chemicals at a low yield. This is what has led to the production of a non-photosynthetic bacterium, *Moorella (M.) thermoacetica* can produce 4 equivalents fused Cysteine molecules (CySS), 2 moles of water (H₂O), and 1 mole of acetic acid by utilizing two moles of CO₂, 8 equivalents of Cysteine, and 8 photons of light.

There are organisms that have the natural ability to utilize CO₂. Typically there is a process that involves multiple steps, CO₂ is transformed into a new chemical that is used by the organism for another pathway. The problem with these organisms is that there isn't much scientific utilization of the product, so research has used organisms that would not use the product due to it not naturally being found within them, non-photosynthetic. In the recent advancements in microbial engineering, scientists, Kelsey Sakimoto; Andrew Wong; and Peidong Yang, were able to improve on this process in an organism called *Moorella thermoacetica* by using a hybrid system. The mechanism being utilized is the manipulation of *M. thermoacetica*, to become photosynthetic via incorporation of the nanoparticles, Cds. Cds functions as a semiconductor that

allows for *M. thermoacetica* to become photosynthetic, *M. thermoacetica*-Cds. With the introduction of Cadmium (Cd^{2+}) into the medium, *M. thermoacetica* is able to fix a sulfur to the ion to create CdS. Studies have shown Cds to be secreted as tiny, light-sensitive nanoparticles that allow for energy absorption. CdS, are useful for their ability to effectively transform energy.

The idea that is particularly vital to the success of these experiments is that *M. thermoacetica* is not typically able to use photosynthesis. Organisms that naturally fix CO_2 typically produce a low yield of product. The process to make energy from *M. thermoacetica* is done by utilizing a known pathway that turn CO_2 into acetic acid. Acetic acid has significant potential to be used for synthesis and extraction due to its utilization for alcohol synthesis. The key point of the Sakimoto study is that they found a method to make acetic acid in high quantity, about 90% yield. This compared to previous microbial engineered organism were about 20% is a significant increase. This system produced acetic acid as waste and is not depleted after synthesis. Acetic acid is made more effective by increasing cysteine available that the organism requires to make the product and by regulating available light levels for the photosynthetic reaction.

The Sakimoto paper gives a great introduction to the potential future of hybrid organisms and their likeliness to be used for synthesis of other molecules. As the future of microbial engineering progresses, they are effective in their ability to produce products with the low cost of starter bacteria and affordable materials like Cadmium and cysteine exhibit the cost effective manner of the process. Bioengineering is certainly one of the greenest ways to make chemicals. With continual expansion, hybrid engineering can be mixed and matched to create an important tool for chemistry that aides in the understanding of biological systems that will allow for more efficient utilization in the future.

Works cited:

Appel, A. M., Bercaw, J. E., Bocarsly, A. B., Dobbek, H., DuBois, D. L., Dupuis, M., ... & Kerfeld, C. A. (2013). Frontiers, opportunities, and challenges in biochemical and chemical catalysis of CO_2 fixation. *Chemical reviews*, 113(8), 6621-6658.

Sakimoto, K. K., Wong, A. B., & Yang, P. (2016). Self-photosensitization of non-photosynthetic bacteria for solar-to-chemical production. *Science*, 351(6268), 74-77. **2021-2022**

Degree Map: Gerontological Studies and Services (BS)

Department of Health and Physical Education
School of Health Sciences & Professional Programs | York College | CUNY
Room HP-203 | (718) 262-2607

The following is a suggested plan of study for completion of this degree program. The goal of a degree map is to ensure that students graduate with no more than 120 credits and in four years.

- All students should speak with an academic advisor about their academic programs. This document is not a substitute for academic advisement.
- Students are encouraged to take Winter and Summer courses to facilitate their progress towards graduation.
- Transfer students do not need to take all courses in the plan; they should consult with an academic advisor.

Course	Credit	Course	Credit
First Year – Fall	15	First Year – Spring	15
English Composition (EC): English 125	3	English Composition (EC): English 126	3
College Option (CO): Health Education 111	3	Mathematical & Quantitative Reasoning (MQR) course*	3
Life & Physical Science (LPS) course	3	Scientific World (SW): Psychology 102 (prerequisite)	3
World Cultures & Global Issues (WCGI) course	3	Individual & Society (IS) course	3
US Experience in its Diversity (USED) course	3	Health Education 201	2
		Elective (Liberal Arts)	4

Course	Credit	Course	Credit
Second Year – Fall	15	Second Year – Spring	15
Gerontology 101	3	Gerontology 201	3
Creative Expression (CE) course	3	College Option (CO): Foreign Language+	3
College Option (CO): Foreign Language+	3	Health Education 321	3
Health Education 223	2	Social Work 203	3
Health Education 224	1	Physical Education 353	2
Elective (Liberal Arts)	3	Health Education 200	4

Course	Credit	Course	Credit
Third Year—Fall	15	Third Year—Spring	15
Gerontology 210	3	Health Education 331 (WI)	3
Health Education 341	3	Gerontology 301	3
College Option (CO): Writing 303 ^Δ	3	Psychology 280	3
Elective (Liberal Arts)	3	Gerontology elective**	3
Elective (Liberal Arts)	3	Elective (Liberal Arts)	3

Course	Credit	Course	Credit
Fourth Year—Fall	15	Fourth Year—Spring	15
Health Education 471 and 473	3	Health Education 472 and 474	3
Social Work 300	3	Health Education 481	3
Psychology 283	3	One additional Flexible Core course	3
Gerontology elective**	3	Elective (Liberal Arts)	3
Elective (Liberal Arts)	3	Elective (Liberal Arts)	3

- ~~York students are required to complete (pass) three (3) Writing intensive (WI) courses: two (2) in the lower division (100-200 level) and one (1) in the upper division (300 level).~~
- ~~If you transferred to York College with credit for General Education requirements, you must still complete one (1) WI course in the upper division within your major. If your major has no upper division WI course, you must take a WI course chosen in consultation with your major advisor.~~
- ~~B.S. students must complete 60 credits of liberal arts—See Bulletin.~~

~~*Students planning to take HE365 to fulfill one Gerontology elective should take MATH111 for their Mathematical & Quantitative Reasoning (MQR) course.~~

~~+Foreign Language courses are determined through placement by the World Languages Department, Room 3C08.~~

~~ΔWRIT303 is a required course for the Gerontological Studies and Services major, and also fulfills a portion of the College Option requirements~~

~~**Students in the Gerontological Studies and Services major are required to complete two (6 crs. total) of the following Gerontology elective courses:~~

~~ANTH302, BIO283, HE365, PE322, POL273, PSY381, SOC236~~